Exercises: Constrained Optimization Methods – Answer Key

Mathematical Foundations for Machine Learning

Projection onto Convex Sets

Exercise 1.1 [Easy]
Let $C = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\}$ be the unit Euclidean ball.

(a) Write down the optimization problem that defines the projection $P_C(z)$ of a point $z \in \mathbb{R}^n$ onto C.

(b) Derive the closed-form expression for $P_C(z)$ by considering the two cases $\|z\|_2 \leq 1$ and $\|z\|_2 > 1$ separately.

(c) Verify that $P_C(z)$ is non-expansive, i.e. $\|P_C(x) - P_C(y)\|_2 \leq \|x - y\|_2$, for the special case $\|x\|_2 > 1$ and $\|y\|_2 \leq 1$.

Solution

(a) The projection is defined as:

$$P_C(z) = \arg\min_{x \in C} \|z - x\|_2^2 = \arg\min_{\|x\|_2 \leq 1} \|z - x\|_2^2.$$

(b) *Case 1:* If $\|z\|_2 \leq 1$, then $z \in C$ and the minimum of $\|z - x\|_2^2$ is 0, attained at $x = z$. So $P_C(z) = z$.
Case 2: If $\|z\|_2 > 1$, the closest point on the ball must lie on the boundary $\|x\|_2 = 1$ (since moving toward z always decreases the distance). The point on the unit sphere closest to z is $x = z/\|z\|_2$ (the radial projection). So $P_C(z) = z/\|z\|_2$.
In summary:

$$P_C(z) = \begin{cases} z & \text{if } \|z\|_2 \leq 1, \\ \frac{z}{\|z\|_2} & \text{if } \|z\|_2 > 1. \end{cases}$$

(c) Let $\|x\|_2 > 1$ and $\|y\|_2 \leq 1$. Then $P_C(x) = x/\|x\|_2$ and $P_C(y) = y$.

$$\|P_C(x) - P_C(y)\|_2 = \left\| \frac{x}{\|x\|_2} - y \right\|_2.$$

Since $x/\|x\|_2$ lies on the segment from 0 to x and $\|x/\|x\|_2 - x\| = \|x\|_2 - 1 \geq 0$, we can write:

$$\left\| \frac{x}{\|x\|_2} - y \right\|_2 \leq \|x - y\|_2 - (\|x\|_2 - 1) + \|y - y\|_2.$$

More directly, note that $x/\|x\|_2$ is the point on the line segment $[0, x]$ closest to the ball. Since y is inside the ball and $x/\|x\|_2$ is closer to y than x is (contracting toward the origin reduces distance to any interior point), we get $\|P_C(x) - P_C(y)\|_2 \leq \|x - y\|_2$.
A clean proof: define $g(t) = \|tx - y\|_2^2$ for $t \in [0, 1]$. We need $g(1/\|x\|_2) \leq g(1)$ when $\|x\|_2 > 1$. Compute $g'(t) = 2t\|x\|_2^2 - 2\langle x, y\rangle$. At $t = 1$: $g'(1) = 2(\|x\|_2^2 - \langle x, y\rangle) \geq 2(\|x\|_2^2 - \|x\|_2) > 0$. Since g is convex in t and $1/\|x\|_2 < 1$, we get $g(1/\|x\|_2) \leq g(1)$.

Exercise 1.2 [Medium]
Let $C = \{x \in \mathbb{R}^n \mid Ax = b\}$ where $A \in \mathbb{R}^{p \times n}$ with $\text{rank}(A) = p < n$.

(a) Using the method of Lagrange multipliers, derive that the projection of a point $z \in \mathbb{R}^n$ onto C is given by

$$P_C(z) = z + A^T(AA^T)^{-1}(b - Az).$$

(b) Show that the projection matrix $P = I - A^T(AA^T)^{-1}A$ is idempotent, i.e. $P^2 = P$, and symmetric.

(c) Verify that for any $z \in \mathbb{R}^n$, the residual $z - P_C(z)$ is orthogonal to the affine subspace C in the sense that $\langle z - P_C(z), v\rangle = 0$ for all $v \in \ker(A)$.

Solution

(a) We solve $\min_x \|z - x\|_2^2$ subject to $Ax = b$. The Lagrangian is:

$$\mathcal{L}(x, \nu) = \|z - x\|_2^2 + \nu^T(Ax - b).$$

Stationarity: $\nabla_x \mathcal{L} = -2(z - x) + A^T\nu = 0 \implies x = z - \frac{1}{2}A^T\nu$.
Substitute into constraint: $A(z - \frac{1}{2}A^T\nu) = b \implies Az - \frac{1}{2}AA^T\nu = b$.
Solve for ν: $\nu = 2(AA^T)^{-1}(Az - b)$.
Plug back: $x = z - A^T(AA^T)^{-1}(Az - b) = z + A^T(AA^T)^{-1}(b - Az)$.
(b) Let $P = I - A^T(AA^T)^{-1}A$.
Idempotent:

$$\begin{aligned} P^2 &= (I - A^T(AA^T)^{-1}A)(I - A^T(AA^T)^{-1}A) \\ &= I - 2A^T(AA^T)^{-1}A + A^T(AA^T)^{-1}\underbrace{AA^T(AA^T)^{-1}}_{=I}A \\ &= I - 2A^T(AA^T)^{-1}A + A^T(AA^T)^{-1}A = P. \end{aligned}$$

Symmetric: $(A^T(AA^T)^{-1}A)^T = A^T((AA^T)^{-1})^TA = A^T((AA^T)^T)^{-1}A = A^T(AA^T)^{-1}A$, since AA^T is symmetric. Hence $P^T = P$.
(c) The residual is $z - P_C(z) = A^T(AA^T)^{-1}(Az - b)$. For any $v \in \ker(A)$, we have $Av = 0$, so:

$$\langle z - P_C(z), v\rangle = (Az - b)^T(AA^T)^{-1}Av = (Az - b)^T(AA^T)^{-1} \cdot 0 = 0.$$

This confirms that the residual is orthogonal to the feasible directions of the affine subspace.

Exercise 1.3 [Medium]
Let $C = \{x \in \mathbb{R}^n \mid x_i \geq 0,\ \sum_{i=1}^n x_i = 1\}$ be the probability simplex.

(a) Show that C is a convex set.

(b) A point $z = (0.5, -0.3, 0.2, 0.9)^T \in \mathbb{R}^4$ needs to be projected onto the simplex $C \subset \mathbb{R}^4$. Formulate the KKT conditions for the projection problem $\min_{x \in C} \|z - x\|_2^2$.

(c) Using the KKT conditions, argue that the solution has the form $x_i = \max\{z_i - \mu, 0\}$ for some $\mu \in \mathbb{R}$, and find the projection $P_C(z)$.

Solution

(a) Let $x, y \in C$ and $\lambda \in [0,1]$. Define $w = \lambda x + (1-\lambda)y$.
$w_i = \lambda x_i + (1-\lambda)y_i \geq 0$ since $x_i, y_i \geq 0$.
$\sum_i w_i = \lambda \sum_i x_i + (1-\lambda)\sum_i y_i = \lambda \cdot 1 + (1-\lambda)\cdot 1 = 1$.
So $w \in C$, proving convexity.
(b) The problem is:

$$\min_x \sum_{i=1}^{4} (z_i - x_i)^2 \quad \text{s.t.} \quad \sum_i x_i = 1,\ -x_i \leq 0\ \forall i.$$

Lagrangian: $\mathcal{L} = \sum_i (z_i - x_i)^2 + \mu(\sum_i x_i - 1) + \sum_i \lambda_i(-x_i)$.
KKT conditions:

1. Stationarity: $-2(z_i - x_i) + \mu - \lambda_i = 0$ for all i.
2. Primal feasibility: $\sum_i x_i = 1$, $x_i \geq 0$.
3. Dual feasibility: $\lambda_i \geq 0$.
4. Complementary slackness: $\lambda_i x_i = 0$ for all i.

(c) From stationarity: $x_i = z_i - \mu/2 + \lambda_i/2$.
If $x_i > 0$: complementary slackness gives $\lambda_i = 0$, so $x_i = z_i - \mu/2$.
If $x_i = 0$: then $\lambda_i = 2(\mu/2 - z_i) \geq 0$, so $z_i \leq \mu/2$.
Letting $\tau = \mu/2$, we get $x_i = \max\{z_i - \tau, 0\}$.
Apply $\sum_i x_i = 1$: Try τ such that $\sum_i \max\{z_i - \tau, 0\} = 1$.
Sort: $z_{(1)} = 0.9, z_{(2)} = 0.5, z_{(3)} = 0.2, z_{(4)} = -0.3$.
Try all active: $\sum(z_i - \tau) = (0.5 - 0.3 + 0.2 + 0.9) - 4\tau = 1.3 - 4\tau = 1 \implies \tau = 0.075$.
Check: $x = (0.425, -0.375, 0.125, 0.825)$. Negative entry, so not all active.
Drop $z_4 = -0.3$: $(0.5 + 0.2 + 0.9) - 3\tau = 1 \implies \tau = 0.2$. Check: $x = (0.3, 0, 0, 0.7)$. But $x_3 = 0.2 - 0.2 = 0$, consistent.
So $P_C(z) = (0.3, 0, 0, 0.7)^T$.

Exercise 1.4 [Hard]
Let $C \subseteq \mathbb{R}^n$ be a closed convex set and P_C the projection operator onto C.

(a) Prove the variational characterization: $x^* = P_C(z)$ if and only if

$$\langle y - x^*, z - x^* \rangle \leq 0 \quad \forall\, y \in C.$$

(Hint: Consider the function $g(t) = \|z - (x^* + t(y - x^*))\|_2^2$ for $t \in [0,1]$.)

(b) Using part (a), prove that the projection is firmly non-expansive:

$$\|P_C(x) - P_C(y)\|_2^2 \leq \langle P_C(x) - P_C(y),\, x - y \rangle.$$

(c) Deduce from (b) that P_C is non-expansive: $\|P_C(x) - P_C(y)\|_2 \leq \|x - y\|_2$.

Solution

(a) ($\Rightarrow$) Let $x^* = P_C(z)$. For any $y \in C$, the point $x^* + t(y - x^*) \in C$ for $t \in [0, 1]$ by convexity. Define $g(t) = \|z - x^* - t(y - x^*)\|_2^2$. Since x^* minimizes over C, we need $g(0) \leq g(t)$ for all $t \in [0, 1]$, so $g'(0) \geq 0$.
$g'(t) = -2\langle y - x^*, z - x^* - t(y - x^*)\rangle$.
$g'(0) = -2\langle y - x^*, z - x^*\rangle \geq 0$, giving $\langle y - x^*, z - x^*\rangle \leq 0$.
($\Leftarrow$) Suppose $\langle y - x^*, z - x^*\rangle \leq 0$ for all $y \in C$. For any $y \in C$:

$$\|z - y\|^2 = \|z - x^* + x^* - y\|^2 = \|z - x^*\|^2 + 2\langle z - x^*, x^* - y\rangle + \|x^* - y\|^2.$$

The middle term: $2\langle z - x^*, x^* - y\rangle = -2\langle y - x^*, z - x^*\rangle \leq 0$ is actually ≥ 0 (sign flip from hypothesis). Wait: $\langle y - x^*, z - x^*\rangle \leq 0$ means $\langle z - x^*, x^* - y\rangle = -\langle z - x^*, y - x^*\rangle \geq 0$. So $\|z - y\|^2 \geq \|z - x^*\|^2 + \|x^* - y\|^2 \geq \|z - x^*\|^2$, proving x^* is the minimizer.
(b) Let $a = P_C(x)$, $b = P_C(y)$. From part (a):
With $z = x$, $x^* = a$: $\langle b - a, x - a\rangle \leq 0$, i.e. $\langle a - b, x - a\rangle \geq 0$.
With $z = y$, $x^* = b$: $\langle a - b, y - b\rangle \leq 0$, i.e. $\langle b - a, y - b\rangle \geq 0$.
Add:

$$\langle a - b, x - a\rangle + \langle b - a, y - b\rangle \geq 0.$$

$$\langle a - b, x - a - y + b\rangle \geq 0.$$

$$\langle a - b, (x - y) - (a - b)\rangle \geq 0.$$

$$\langle a - b, x - y\rangle \geq \|a - b\|^2.$$

This is exactly $\|P_C(x) - P_C(y)\|^2 \leq \langle P_C(x) - P_C(y), x - y\rangle$.
(c) Apply Cauchy–Schwarz to the right side of (b):

$$\|P_C(x) - P_C(y)\|^2 \leq \langle P_C(x) - P_C(y), x - y\rangle \leq \|P_C(x) - P_C(y)\| \cdot \|x - y\|.$$

Divide both sides by $\|P_C(x) - P_C(y)\|$ (if zero, the result is trivial):

$$\|P_C(x) - P_C(y)\| \leq \|x - y\|.$$

Projected Gradient Descent

Exercise 2.1 [Easy]
Consider the constrained optimization problem

$$\min_{x \in C} f(x) = \frac{1}{2}x^T Q x - c^T x,$$

where $Q = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$, $c = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, and $C = \{x \in \mathbb{R}^2 \mid x_1 \geq 0,\ x_2 \geq 0\}$ is the non-negative orthant.

(a) Compute the gradient $\nabla f(x)$ and the Lipschitz constant L of the gradient.

(b) Starting from $x^{(0)} = (0, 0)^T$ with step size $t = 1/L$, perform two iterations of projected gradient descent:

$$x^{(k+1)} = P_C\big(x^{(k)} - t\,\nabla f(x^{(k)})\big).$$

(c) What is the unconstrained minimizer? Is it equal to the constrained minimizer? Why or why not?

Solution

(a) $\nabla f(x) = Qx - c = \begin{pmatrix} 2x_1 - 1 \\ 4x_2 - 2 \end{pmatrix}$.
The Lipschitz constant of the gradient is $L = \|Q\|_2 = \lambda_{\max}(Q) = 4$.
(b) Step size $t = 1/4$.
Iteration 1: $\nabla f(x^{(0)}) = (-1, -2)^T$. $x^{(0)} - t\nabla f(x^{(0)}) = (0, 0) - \frac{1}{4}(-1, -2) = (0.25, 0.5)^T$. Since both components are non-negative: $x^{(1)} = P_C(0.25, 0.5) = (0.25, 0.5)^T$.
Iteration 2: $\nabla f(x^{(1)}) = (2(0.25)-1, 4(0.5)-2) = (-0.5, 0)^T$. $x^{(1)} - t\nabla f(x^{(1)}) = (0.25, 0.5) - \frac{1}{4}(-0.5, 0) = (0.375, 0.5)^T$. $x^{(2)} = P_C(0.375, 0.5) = (0.375, 0.5)^T$.
(c) Unconstrained minimizer: $\nabla f = 0 \implies x^* = Q^{-1}c = (0.5, 0.5)^T$. Since $x_1^* = 0.5 > 0$ and $x_2^* = 0.5 > 0$, the unconstrained minimizer lies in C, so it equals the constrained minimizer. The constraint is inactive.

Exercise 2.2 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex with L-Lipschitz continuous gradient, and let C be a closed convex set. Consider projected gradient descent with constant step size $t = 1/L$.

(a) State the optimality condition for $x^* = \arg\min_{x \in C} f(x)$ in terms of the projection P_C and the gradient ∇f.

(b) Show that if $x^* = P_C(x^* - t\,\nabla f(x^*))$, then $\langle \nabla f(x^*), y - x^* \rangle \geq 0$ for all $y \in C$. Interpret this geometrically.

(c) Let $f(x) = \|x - p\|_2^2$ for some $p \notin C$. Show that projected gradient descent with step size $t = 1/L$ converges to the projection $P_C(p)$ in a single step from any $x^{(0)} \in C$. What is L in this case?

Solution

(a) The optimality condition is:

$$x^* = P_C\big(x^* - t\,\nabla f(x^*)\big) \quad \text{for any } t > 0.$$

Equivalently, $P_C(x^* - \frac{1}{L}\nabla f(x^*)) = x^*$.
(b) If $x^* = P_C(x^* - t\nabla f(x^*))$, then by the variational characterization of projections, for all $y \in C$:

$$\langle y - x^*,\ (x^* - t\nabla f(x^*)) - x^* \rangle \leq 0.$$

$$\langle y - x^*,\ -t\nabla f(x^*) \rangle \leq 0.$$

Since $t > 0$: $\langle \nabla f(x^*),\ y - x^* \rangle \geq 0$ for all $y \in C$.
Geometric interpretation: The negative gradient $-\nabla f(x^*)$ makes an obtuse angle with every feasible direction $y - x^*$. This means we cannot move in any feasible direction to decrease f.
(c) Here $f(x) = \|x - p\|_2^2$, so $\nabla f(x) = 2(x - p)$ and the Hessian is $2I$, giving $L = 2$.
With $t = 1/L = 1/2$:

$$x^{(1)} = P_C\left(x^{(0)} - \frac{1}{2} \cdot 2(x^{(0)} - p)\right) = P_C\left(x^{(0)} - x^{(0)} + p\right) = P_C(p).$$

Thus regardless of $x^{(0)}$, we get $x^{(1)} = P_C(p)$, which is the exact solution. The step size $1/L$ perfectly cancels the quadratic curvature, reducing the problem to a single projection.

Exercise 2.3 [Hard]

Consider the constrained LASSO problem:

$$\min_{x\in\mathbb{R}^n} \frac{1}{2}\|Ax-b\|_2^2 \quad \text{s.t.} \quad \|x\|_1 \le \tau,$$

where $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and $\tau > 0$.

(a) Show that the objective $f(x) = \frac{1}{2}\|Ax-b\|_2^2$ has Lipschitz continuous gradient with constant $L = \|A^TA\|_2 = \sigma_{\max}(A)^2$.

(b) Write down the projected gradient descent update for this problem. Which subproblem needs to be solved at each step?

(c) The projection onto the ℓ_1-ball $C = \{x \mid \|x\|_1 \le \tau\}$ can be computed via soft-thresholding: $(P_C(z))_i = \operatorname{sign}(z_i)\max\{|z_i|-\lambda, 0\}$ where $\lambda \ge 0$ is chosen so that $\|P_C(z)\|_1 = \tau$ (or $\lambda = 0$ if $\|z\|_1 \le \tau$). Prove that this operator is well-defined, i.e. that there exists a unique $\lambda \ge 0$ such that $\sum_i \max\{|z_i|-\lambda, 0\} = \tau$ whenever $\|z\|_1 > \tau$.

(d) State the convergence rate of projected gradient descent for this problem and compare it to the accelerated variant.

Solution

(a) $\nabla f(x) = A^T(Ax-b)$. For any x, y:

$$\|\nabla f(x) - \nabla f(y)\| = \|A^TA(x-y)\| \le \|A^TA\|_2\|x-y\|.$$

Since $\|A^TA\|_2 = \lambda_{\max}(A^TA) = \sigma_{\max}(A)^2$, we get $L = \sigma_{\max}(A)^2$.

(b) The update is:

$$x^{(k+1)} = P_C\left(x^{(k)} - \frac{1}{L}A^T(Ax^{(k)}-b)\right),$$

where $C = \{x : \|x\|_1 \le \tau\}$. The subproblem at each step is: compute a gradient step, then project the result onto the ℓ_1-ball of radius τ.

(c) Define $h(\lambda) = \sum_{i=1}^n \max\{|z_i|-\lambda, 0\}$ for $\lambda \ge 0$.
h is continuous (as a sum of continuous functions).
h is strictly decreasing on $[0, \max_i |z_i|]$: increasing λ reduces each term.
$h(0) = \|z\|_1 > \tau$ by assumption.
$h(\max_i |z_i|) = 0 < \tau$.
By the intermediate value theorem, there exists $\lambda^* \in (0, \max_i |z_i|)$ with $h(\lambda^*) = \tau$. Uniqueness follows from strict monotonicity (where h is positive). More precisely, h is piecewise linear and non-increasing, and strictly decreasing until it reaches 0, so the root is unique.

(d) Projected gradient descent achieves:

$$f(x^{(k)}) - f^* \le \frac{L\|x^{(0)}-x^*\|^2}{2k}.$$

This is $O(1/k)$, so to achieve ε-accuracy we need $O(L/\varepsilon)$ iterations.
The accelerated projected gradient descent (Nesterov) achieves:

$$f(x^{(k)}) - f^* \le \frac{2L\|x^{(0)}-x^*\|^2}{(k+1)^2},$$

which is $O(1/k^2)$, requiring only $O(\sqrt{L/\varepsilon})$ iterations — a quadratic improvement.

Interior Point / Barrier Method

Exercise 3.1 [Easy]
Consider the linear program

$$\min_{x\in\mathbb{R}^2} c^T x \quad \text{s.t.} \quad x_1 \geq 0,\ x_2 \geq 0,\ x_1 + x_2 \leq 1,$$

with $c = (1, 2)^T$.

(a) Write down the logarithmic barrier function $\phi(x)$ for the inequality constraints.

(b) Formulate the barrier problem: $\min_x\ t\,c^T x + \phi(x)$ (there are no equality constraints here).

(c) For a given $t > 0$, find the central point $x^*(t)$ by solving the first-order optimality conditions of the barrier problem.

(d) Verify that $x^*(t) \to (0, 0)^T$ (the LP optimum) as $t \to \infty$. What is the duality gap $f(x^*(t)) - p^*$ in terms of t?

Solution

(a) The constraints rewritten as $g_i(x) \leq 0$: $g_1 = -x_1$, $g_2 = -x_2$, $g_3 = x_1 + x_2 - 1$.

$$\phi(x) = -\log(x_1) - \log(x_2) - \log(1 - x_1 - x_2).$$

(b) The barrier problem is:

$$\min_{x_1,x_2}\ t(x_1 + 2x_2) - \log(x_1) - \log(x_2) - \log(1 - x_1 - x_2).$$

(c) Setting partial derivatives to zero:

$$\frac{\partial}{\partial x_1} : t - \frac{1}{x_1} + \frac{1}{1 - x_1 - x_2} = 0,$$

$$\frac{\partial}{\partial x_2} : 2t - \frac{1}{x_2} + \frac{1}{1 - x_1 - x_2} = 0.$$

Subtracting: $-t + \frac{1}{x_2} - \frac{1}{x_1} = 0$, so $\frac{1}{x_1} = \frac{1}{x_2} - t$.
From the structure, let $s = 1 - x_1 - x_2$. The first equation gives $tx_1 s = s - x_1$, i.e. $\frac{1}{x_1} = t + \frac{1}{s}$, and $\frac{1}{x_2} = 2t + \frac{1}{s}$.
Define dual variables $\lambda_i^*(t) = \frac{1}{tx_i}$ for $i = 1, 2$ and $\lambda_3^*(t) = \frac{1}{ts}$.
From the equations: $x_1 = \frac{1}{t+1/s}$, $x_2 = \frac{1}{2t+1/s}$, and we solve for s from $s = 1 - x_1 - x_2$. For large t: $x_1 \approx 1/t$, $x_2 \approx 1/(2t)$, $s \approx 1 - 3/(2t)$.
(d) As $t \to \infty$: $x_1^*(t) \to 0$, $x_2^*(t) \to 0$, so $x^*(t) \to (0, 0)$, which is the LP optimum with $p^* = 0$.
The duality gap is $f(x^*(t)) - p^* \leq m/t = 3/t$ (since there are $m = 3$ inequality constraints).

Exercise 3.2 [Medium]
Consider the general barrier method for the problem

$$\min_x f(x) \quad \text{s.t.} \quad g_i(x) \leq 0,\ i = 1, \ldots, m, \quad Ax = b.$$

(a) Define the logarithmic barrier function $\phi(x)$ and write the barrier subproblem for parameter $t > 0$.

(b) Let $x^*(t)$ be the solution of the barrier subproblem. Show that the KKT conditions of the barrier subproblem yield dual-feasible points for the original problem. Specifically, define $\lambda_i^*(t) = -\frac{1}{t\, g_i(x^*(t))}$ and show $\lambda_i^*(t) \geq 0$.

(c) Show that $f(x^*(t)) - p^* \leq \frac{m}{t}$, where p^* is the optimal value of the original problem.

(d) Interpret the complementary slackness violation: show $\lambda_i^*(t) \cdot (-g_i(x^*(t))) = \frac{1}{t}$ and explain what happens as $t \to \infty$.

Solution

(a) $\phi(x) = -\sum_{i=1}^m \log(-g_i(x))$. The barrier subproblem is:

$$\min_x \; t\, f(x) + \phi(x) \quad \text{s.t.} \quad Ax = b.$$

(b) The optimality condition for $x^*(t)$ (equality-constrained problem) gives:

$$0 = t\, \nabla f(x^*(t)) + \nabla \phi(x^*(t)) + A^T \hat{\mu},$$

where $\nabla \phi(x^*(t)) = -\sum_{i=1}^m \frac{1}{g_i(x^*(t))} \nabla g_i(x^*(t)) = \sum_{i=1}^m \frac{-1}{g_i(x^*(t))} \nabla g_i(x^*(t))$.
Define $\lambda_i^*(t) = \frac{-1}{t\, g_i(x^*(t))}$. Since $x^*(t)$ is strictly feasible ($g_i(x^*(t)) < 0$) and $t > 0$, we have $\lambda_i^*(t) > 0$.
Dividing the optimality condition by t:

$$\nabla f(x^*(t)) + \sum_i \lambda_i^*(t) \nabla g_i(x^*(t)) + A^T \mu^*(t) = 0,$$

where $\mu^*(t) = \hat{\mu}/t$. This is precisely the stationarity condition of the Lagrangian of the original problem, so $(\lambda^*(t), \mu^*(t))$ is dual feasible.
(c) The dual function evaluated at $(\lambda^*(t), \mu^*(t))$ satisfies (since $x^*(t)$ minimizes the Lagrangian by stationarity):

$$q(\lambda^*(t), \mu^*(t)) = f(x^*(t)) + \sum_i \lambda_i^*(t) g_i(x^*(t)) + \langle \mu^*(t), Ax^*(t) - b \rangle.$$

Since $Ax^*(t) = b$ and $\lambda_i^*(t) g_i(x^*(t)) = \frac{-1}{t\, g_i} \cdot g_i = -1/t$:

$$q(\lambda^*, \mu^*) = f(x^*(t)) - m/t.$$

By weak duality $p^* \geq q(\lambda^*, \mu^*)$, so $f(x^*(t)) - p^* \leq f(x^*(t)) - q(\lambda^*, \mu^*) = m/t$.
(d) $\lambda_i^*(t) \cdot (-g_i(x^*(t))) = \frac{-1}{t\, g_i} \cdot (-g_i) = \frac{1}{t}$.
Exact complementary slackness requires $\lambda_i g_i = 0$, but here the product is $1/t$ instead. As $t \to \infty$, this violation vanishes: $\lambda_i^*(t) g_i(x^*(t)) \to 0$, so the KKT conditions of the original problem are approximately satisfied, with accuracy controlled by t.

Exercise 3.3 [Hard]
Consider the inequality-constrained quadratic program:

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^T Q x + c^T x \quad \text{s.t.} \quad a_i^T x \leq b_i, \; i = 1, \ldots, m,$$

where $Q \succ 0$.

(a) Write the barrier subproblem for parameter $t > 0$ and compute its gradient and Hessian with respect to x.

(b) Suppose $n = 1$, $Q = (2)$, $c = (-3)$, with a single constraint $x \leq 2$. Solve for the central path $x^*(t)$ explicitly as a function of t, and verify that $x^*(t) \to 3/2$ as $t \to \infty$.

(c) In the barrier method, one increases t by a factor $\gamma > 1$ at each outer iteration. Explain the trade-off in choosing γ: what happens for very small γ vs. very large γ?

(d) For the general case, explain how Newton's method would be applied to solve each barrier subproblem. Write down the Newton step Δx in terms of the Hessian and gradient of the barrier objective.

Solution

(a) The barrier objective is:

$$B_t(x) = t\left(\frac{1}{2}x^T Qx + c^T x\right) - \sum_{i=1}^{m} \log(b_i - a_i^T x).$$

Gradient:

$$\nabla B_t(x) = t(Qx + c) + \sum_{i=1}^{m} \frac{a_i}{b_i - a_i^T x}.$$

Hessian:

$$\nabla^2 B_t(x) = tQ + \sum_{i=1}^{m} \frac{a_i a_i^T}{(b_i - a_i^T x)^2}.$$

Note: $\nabla^2 B_t(x) \succ 0$ since $tQ \succ 0$, so each subproblem is strictly convex.
(b) With $n = 1$, $Q = 2$, $c = -3$, constraint $x \leq 2$:

$$B_t(x) = t(x^2 - 3x) - \log(2 - x).$$

$$B_t'(x) = t(2x - 3) + \frac{1}{2 - x} = 0.$$

Let $s = 2 - x$: $t(2(2 - s) - 3) + 1/s = 0 \implies t(1 - 2s) + 1/s = 0$. $t - 2ts + 1/s = 0 \implies ts^2 - ts - 1/(2) = 0$... Let us solve directly: $t(2x - 3) = -\frac{1}{2-x}$, so $t(2x - 3)(2 - x) = -1$. Expand: $t(4x - 2x^2 - 6 + 3x) = -1$, i.e. $t(-2x^2 + 7x - 6) = -1$. $2tx^2 - 7tx + 6t - 1 = 0$. Using the quadratic formula: $x = \frac{7t \pm \sqrt{49t^2 - 8t(6t-1)}}{4t} = \frac{7t \pm \sqrt{t^2+8t}}{4t}$.
Since $x < 2$ (feasibility) and x should approach $3/2$ from below: $x^*(t) = \frac{7t - \sqrt{t^2+8t}}{4t} = \frac{7}{4} - \frac{\sqrt{t^2+8t}}{4t}$.
As $t \to \infty$: $\sqrt{t^2 + 8t}/t = \sqrt{1 + 8/t} \to 1$, so $x^*(t) \to 7/4 - 1/4 = 6/4 = 3/2$.
This matches the unconstrained minimizer $x^* = 3/2$ (since $2(3/2) - 3 = 0$ and $3/2 < 2$).
(c) *Small γ (close to 1):* Each new subproblem changes little from the previous one, so the previous central point is a warm start close to the new optimum — few Newton steps per centering. However, many outer iterations are needed to reach a large t.
Large γ: Fewer outer iterations to reach large t, but each centering step starts far from the new optimum, requiring many Newton steps.
In practice, $\gamma \in [3, 100]$ gives roughly constant total Newton steps.
(d) Newton's method applied to the barrier subproblem computes:

$$\Delta x = -\left(\nabla^2 B_t(x)\right)^{-1} \nabla B_t(x)$$

$$= -\left(tQ + \sum_{i=1}^{m} \frac{a_i a_i^T}{(b_i - a_i^T x)^2}\right)^{-1} \left(t(Qx + c) + \sum_{i=1}^{m} \frac{a_i}{b_i - a_i^T x}\right).$$

The update is $x \leftarrow x + \Delta x$ (or with backtracking line search $x \leftarrow x + \alpha\, \Delta x$).

Quadratic Programs with Linear Equality Constraints

Exercise 4.1 [Easy]
Consider the equality-constrained quadratic program:

$$\min_{x\in\mathbb{R}^n} \frac{1}{2}x^TQx + c^Tx \quad \text{s.t.} \quad Ax = b,$$

where $Q \succ 0$, $A \in \mathbb{R}^{p\times n}$ with $\text{rank}(A) = p$.

(a) Write down the Lagrangian and derive the KKT system (a linear system in x and the multiplier ν).

(b) Show that the KKT system can be written as

$$\begin{pmatrix} Q & A^T \\ A & 0 \end{pmatrix}\begin{pmatrix} x \\ \nu \end{pmatrix} = \begin{pmatrix} -c \\ b \end{pmatrix}.$$

(c) Solve the system for $Q = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, $c = \begin{pmatrix} -4 \\ -6 \end{pmatrix}$, $A = \begin{pmatrix} 1 & 1 \end{pmatrix}$, $b = 3$.

Solution

(a) Lagrangian: $\mathcal{L}(x,\nu) = \frac{1}{2}x^TQx + c^Tx + \nu^T(Ax - b)$.
Stationarity: $\nabla_x\mathcal{L} = Qx + c + A^T\nu = 0 \implies Qx + A^T\nu = -c$.
Primal feasibility: $Ax = b$.
(b) Combining the two conditions:

$$\begin{pmatrix} Q & A^T \\ A & 0 \end{pmatrix}\begin{pmatrix} x \\ \nu \end{pmatrix} = \begin{pmatrix} -c \\ b \end{pmatrix}.$$

This is the KKT system. Since $Q \succ 0$ and A has full row rank, this matrix is nonsingular, guaranteeing a unique solution.
(c) The system is:

$$\begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ \nu \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \\ 3 \end{pmatrix}.$$

From stationarity: $x_1 = (4-\nu)/2 = 2 - \nu/2$, $x_2 = (6-\nu)/2 = 3 - \nu/2$.
Substitute into $x_1 + x_2 = 3$: $(2 - \nu/2) + (3 - \nu/2) = 3$, so $5 - \nu = 3$, giving $\nu = 2$.
Thus $x_1 = 1$, $x_2 = 2$, $\nu = 2$. The constrained minimizer is $x^* = (1,2)^T$.

Exercise 4.2 [Medium]
Consider the problem of finding the minimum-norm solution:

$$\min_{x\in\mathbb{R}^n} \frac{1}{2}\|x\|_2^2 \quad \text{s.t.} \quad Ax = b,$$

where $A \in \mathbb{R}^{p\times n}$ with $\text{rank}(A) = p$ and $p < n$.

(a) Derive the closed-form solution $x^* = A^T(AA^T)^{-1}b$ from the KKT conditions.

(b) Show that this is the Moore–Penrose pseudoinverse solution: $x^* = A^\dagger b$ where $A^\dagger = A^T(AA^T)^{-1}$ for full row-rank A.

(c) In the context of linear regression with more features than data points ($n > N$), this corresponds to the minimum-norm interpolation. If $A = X \in \mathbb{R}^{N\times n}$ and $b = y \in \mathbb{R}^N$, express the solution $\hat{\theta}$ and explain why this connects to the unregularized least-squares solution in the overparameterized regime.

(d) Show that x^* lies in the row space of A, i.e. $x^* \in \text{range}(A^T)$.

Solution

(a) Lagrangian: $\mathcal{L}(x, \nu) = \frac{1}{2}\|x\|^2 + \nu^T(Ax - b)$.
Stationarity: $x + A^T\nu = 0 \implies x = -A^T\nu$.
Feasibility: $Ax = b \implies A(-A^T\nu) = b \implies -AA^T\nu = b \implies \nu = -(AA^T)^{-1}b$.
Substitute back: $x^* = -A^T(-(AA^T)^{-1}b) = A^T(AA^T)^{-1}b$.
(b) For A with full row rank, the Moore–Penrose pseudoinverse is $A^\dagger = A^T(AA^T)^{-1}$. This can be verified: $AA^\dagger = AA^T(AA^T)^{-1} = I_p$ (left inverse property). So $x^* = A^\dagger b$, confirming it is the pseudoinverse solution.
(c) With $A = X$ and $b = y$: $\hat{\theta} = X^T(XX^T)^{-1}y$.
When $n > N$ (more parameters than data), the system $X\theta = y$ is underdetermined. Among all interpolating solutions ($X\theta = y$), $\hat{\theta}$ is the one with minimum ℓ_2-norm. This is the implicit bias of gradient descent on unregularized least squares in overparameterized models: gradient descent from the origin converges to this minimum-norm solution.
(d) From the derivation, $x^* = A^T(AA^T)^{-1}b = A^Tw$ where $w = (AA^T)^{-1}b \in \mathbb{R}^p$. Thus x^* is a linear combination of the rows of A (i.e. the columns of A^T), so $x^* \in \text{range}(A^T)$.

Exercise 4.3 [Hard]
Consider the equality-constrained quadratic program with a general positive semidefinite matrix:

$$\min_{x \in \mathbb{R}^n} \frac{1}{2}x^TQx + c^Tx \quad \text{s.t.} \quad Ax = b,$$

where $Q \succeq 0$ (PSD but possibly singular), $A \in \mathbb{R}^{p \times n}$, $\text{rank}(A) = p$.

(a) The KKT matrix $K = \begin{pmatrix} Q & A^T \\ A & 0 \end{pmatrix}$ may be singular when Q is only PSD. Give a necessary and sufficient condition on Q and A for K to be nonsingular (Hint: consider $\ker(Q) \cap \ker(A)$).

(b) When K is nonsingular, derive the Schur complement formula: eliminating x from the KKT system yields the reduced system $AQ^{-1}A^T\nu = AQ^{-1}c + b$ (assuming $Q \succ 0$). When is this approach computationally preferable?

(c) Apply part (b) to the Ridge regression problem: $\min_\theta \frac{1}{2}\|X\theta - y\|^2 + \frac{\alpha}{2}\|\theta\|^2$ subject to $e^T\theta = 0$ (zero-mean constraint), where $e = (1, \ldots, 1)^T$. Derive the constrained Ridge estimator.

Solution

(a) The KKT matrix K is nonsingular if and only if $\ker(Q) \cap \ker(A) = \{0\}$.
Proof: Suppose $Kv = 0$ where $v = (x, \nu)$. Then $Qx + A^T\nu = 0$ and $Ax = 0$. Multiplying the first equation by x^T: $x^TQx + x^TA^T\nu = 0$. Since $Ax = 0$, we get $x^TQx = 0$. Since $Q \succeq 0$, this means $Qx = 0$. Then $A^T\nu = 0$, and since A has full row rank, $\nu = 0$. So we need $Qx = 0$ and $Ax = 0$ to imply $x = 0$, which is exactly $\ker(Q) \cap \ker(A) = \{0\}$.
(b) From stationarity: $Qx = -c - A^T\nu \implies x = -Q^{-1}(c + A^T\nu)$ (when $Q \succ 0$).
Substitute into $Ax = b$: $-AQ^{-1}(c+A^T\nu) = b \implies -AQ^{-1}A^T\nu = b+AQ^{-1}c$. $AQ^{-1}A^T\nu = -(b + AQ^{-1}c)$.
This is a $p \times p$ system. When $p \ll n$, solving this reduced system is much cheaper than the full $(n + p) \times (n + p)$ KKT system.
(c) The objective is $f(\theta) = \frac{1}{2}\|X\theta - y\|^2 + \frac{\alpha}{2}\|\theta\|^2 = \frac{1}{2}\theta^T(X^TX + \alpha I)\theta - y^TX\theta + \frac{1}{2}\|y\|^2$.
So $Q = X^TX + \alpha I \succ 0$ and $c = -X^Ty$. Constraint: $A = e^T$, $b = 0$.
From the Schur complement: $e^T(X^TX + \alpha I)^{-1}e \cdot \nu = -e^T(X^TX + \alpha I)^{-1}X^Ty$.
Let $M = (X^TX + \alpha I)^{-1}$. Then $\nu = -\frac{e^TMX^Ty}{e^TMe}$.
The unconstrained Ridge solution is $\hat{\theta}_{\text{ridge}} = MX^Ty$.
The constrained solution is:

$$\hat{\theta}_{\text{constrained}} = \hat{\theta}_{\text{ridge}} - \frac{e^T\hat{\theta}_{\text{ridge}}}{e^TMe}Me.$$

This subtracts a correction that enforces $e^T\theta = 0$, projecting the Ridge solution onto the constraint.

Penalty Methods and Accelerated Gradient Descent

Exercise 5.1 [Easy]
Consider the equality-constrained problem:

$$\min_{x\in\mathbb{R}^2} x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 = 1.$$

(a) The quadratic penalty method replaces this with the unconstrained problem

$$\min_x f(x) + \frac{\rho}{2}\|h(x)\|^2 = x_1^2 + x_2^2 + \frac{\rho}{2}(x_1 + x_2 - 1)^2.$$

Find the minimizer $x^*(\rho)$ for a given $\rho > 0$.

(b) Show that $x^*(\rho) \to (1/2, 1/2)^T$ as $\rho \to \infty$, and verify that $(1/2, 1/2)^T$ is the constrained minimizer.

(c) Explain why, in practice, one cannot simply use a very large ρ from the start.

Solution

(a) The penalized objective is $P_\rho(x) = x_1^2 + x_2^2 + \frac{\rho}{2}(x_1 + x_2 - 1)^2$.
$\frac{\partial P_\rho}{\partial x_1} = 2x_1 + \rho(x_1 + x_2 - 1) = 0$, $\frac{\partial P_\rho}{\partial x_2} = 2x_2 + \rho(x_1 + x_2 - 1) = 0$.
Subtracting: $2x_1 - 2x_2 = 0 \implies x_1 = x_2$.
Substitute: $2x_1 + \rho(2x_1 - 1) = 0 \implies x_1(2 + 2\rho) = \rho \implies x_1 = \frac{\rho}{2+2\rho}$.
So $x^*(\rho) = \left(\frac{\rho}{2(1+\rho)}, \frac{\rho}{2(1+\rho)}\right)^T$.
(b) As $\rho \to \infty$: $\frac{\rho}{2(1+\rho)} = \frac{1}{2} \cdot \frac{\rho}{1+\rho} \to \frac{1}{2}$. So $x^*(\rho) \to (1/2, 1/2)^T$.
Verification: By Lagrange multipliers on the constrained problem, $\nabla f = \lambda \nabla h$ gives $(2x_1, 2x_2) = \lambda(1, 1)$, so $x_1 = x_2$. With $x_1 + x_2 = 1$: $x^* = (1/2, 1/2)^T$.
(c) As ρ increases, the penalized objective becomes increasingly ill-conditioned: the Hessian has eigenvalues 2 and $2 + 2\rho$, giving condition number $(1 + \rho)$. For very large ρ, gradient-based methods converge extremely slowly due to this poor conditioning. The standard approach is to gradually increase ρ through a sequence $\rho_1 < \rho_2 < \cdots$, warm-starting each subproblem.

Exercise 5.2 [Medium]
Consider Nesterov's accelerated projected gradient descent for $\min_{x \in C} f(x)$ where f has L-Lipschitz continuous gradient and C is closed convex. The iterations are:

$$x^{(k+1)} = P_C\left(y^{(k)} - \frac{1}{L}\nabla f(y^{(k)})\right), \qquad y^{(k+1)} = x^{(k+1)} + \frac{k}{k+3}\left(x^{(k+1)} - x^{(k)}\right),$$

with $y^{(0)} = x^{(0)} \in C$.

(a) Compare the convergence rates of projected gradient descent and accelerated projected gradient descent. Express both in terms of the number of iterations needed to achieve $f(x^{(k)}) - f^* \leq \varepsilon$.

(b) The momentum coefficient is $\beta_k = \frac{k}{k+3}$. What is $\lim_{k\to\infty} \beta_k$? Why does this not cause divergence?

(c) Consider $f(x) = \frac{1}{2}x^T Q x$ with $Q = \mathrm{diag}(1, 100)$ and $C = \mathbb{R}^2$ (unconstrained). Starting from $x^{(0)} = (100, 100)^T$, explain qualitatively why accelerated gradient descent converges faster than standard gradient descent for this problem.

Solution

(a) *Projected gradient descent:*

$$f(x^{(k)}) - f^* \leq \frac{L\|x^{(0)} - x^*\|^2}{2k} \implies k = O\left(\frac{L}{\varepsilon}\right) \text{ iterations for } \varepsilon\text{-accuracy.}$$

Accelerated projected gradient descent:

$$f(x^{(k)}) - f^* \leq \frac{2L\|x^{(0)} - x^*\|^2}{(k+1)^2} \implies k = O\left(\sqrt{\frac{L}{\varepsilon}}\right) \text{ iterations.}$$

The accelerated method achieves a quadratic improvement in the iteration count.
(b) $\lim_{k\to\infty} \beta_k = \lim_{k\to\infty} \frac{k}{k+3} = 1$.
This does not cause divergence because: (1) The momentum coefficient approaches 1 slowly, giving the algorithm time to settle. (2) The step $x^{(k+1)} - x^{(k)}$ itself decreases as the iterates converge, so the momentum term $\beta_k(x^{(k+1)} - x^{(k)})$ shrinks. (3) The projection P_C clips any overshooting back to the feasible set.
(c) The condition number is $\kappa = \lambda_{\max}/\lambda_{\min} = 100$. Standard gradient descent with step size $1/L = 1/100$ makes tiny steps along the x_1-direction (eigenvalue 1) while adequately handling the x_2-direction (eigenvalue 100). This leads to a slow zigzagging convergence.
Accelerated gradient descent uses momentum to accumulate velocity along the slow x_1-direction, effectively "remembering" the consistent gradient direction and accelerating along it. This is particularly beneficial when the condition number is large, as the momentum overcomes the disparity between the eigenvalues. The convergence depends on $\sqrt{L}$ rather than L, reducing the effective condition number from 100 to $\sqrt{100} = 10$.

Exercise 5.3 [Hard]
Consider the inequality-constrained problem:

$$\min_{x\in\mathbb{R}^n} f(x) \quad \text{s.t.} \quad g_i(x) \leq 0,\ i = 1, \ldots, m.$$

(a) The **exterior penalty method** uses the penalized objective $P_\rho(x) = f(x) + \frac{\rho}{2}\sum_{i=1}^m \max\{0, g_i(x)\}^2$. Explain why, for finite ρ, the minimizer $x^*(\rho)$ typically violates the constraints. Contrast this with the barrier method.

(b) Prove that if f and all g_i are convex, then P_ρ is also convex.

(c) Consider $\min_{x\in\mathbb{R}} x^2$ subject to $x \geq 1$ (equivalently $-x + 1 \leq 0$). Solve the penalty subproblem $\min_x x^2 + \frac{\rho}{2}\max\{0, 1-x\}^2$ analytically, find $x^*(\rho)$, and show $x^*(\rho) \to 1$ as $\rho \to \infty$.

(d) Compare the three approaches for handling constraints: penalty method, barrier method, and projected gradient descent. Discuss when each is most appropriate.

Solution

(a) For finite ρ, the penalty method balances the objective $f(x)$ against the penalty $\frac{\rho}{2}\sum \max\{0, g_i\}^2$. The minimizer finds a trade-off: slightly violating constraints if the reduction in f outweighs the penalty cost. As $\rho \to \infty$, violations become infinitely expensive and $x^*(\rho)$ approaches feasibility.
In contrast, the barrier method uses $\phi(x) = -\sum \log(-g_i(x))$, which is defined only for strictly feasible points ($g_i(x) < 0$). Thus barrier iterates are always strictly feasible, approaching the boundary from the interior.
(b) Each g_i is convex, so $\max\{0, g_i(x)\}$ is convex (pointwise max of convex functions). The square of a non-negative convex function is convex: if $h(x) \geq 0$ is convex, then $h(x)^2$ is convex (since $h'' \cdot h + (h')^2 \geq 0$ when $h \geq 0$ and h is convex). Alternatively, $\max\{0, g_i(x)\}^2$ is the composition of the convex non-decreasing function $t \mapsto t^2$ on $[0, \infty)$ with the convex function $\max\{0, g_i\}$, which preserves convexity.
Thus $P_\rho = f + \frac{\rho}{2}\sum \max\{0, g_i\}^2$ is a sum of convex functions, hence convex.
(c) The penalized problem: $P_\rho(x) = x^2 + \frac{\rho}{2}\max\{0, 1-x\}^2$.
Case 1: $x \geq 1$. $P_\rho(x) = x^2$, minimized at $x = 1$ with value 1.
Case 2: $x < 1$. $P_\rho(x) = x^2 + \frac{\rho}{2}(1-x)^2$. Derivative: $2x - \rho(1-x) = 2x - \rho + \rho x = (2+\rho)x - \rho = 0$. $x = \frac{\rho}{2+\rho}$. This is < 1 for all finite ρ. Value: $\left(\frac{\rho}{2+\rho}\right)^2 + \frac{\rho}{2}\left(\frac{2}{2+\rho}\right)^2 = \frac{\rho^2+2\rho}{(2+\rho)^2} = \frac{\rho}{2+\rho} < 1$.
Since $\frac{\rho}{2+\rho} < 1$, Case 2 gives a lower value, so $x^*(\rho) = \frac{\rho}{2+\rho}$.
As $\rho \to \infty$: $x^*(\rho) = \frac{\rho}{2+\rho} \to 1$. Note $x^*(\rho) < 1$ for all finite ρ, confirming constraint violation.
(d) *Penalty method:* Simple to implement, works for general constraints. Cons: iterates are infeasible, ill-conditioning for large ρ, only approximate feasibility.
Barrier method: Iterates always strictly feasible, principled duality gap bound m/t, well-suited for inequality constraints with smooth g_i. Cons: requires strictly feasible starting point, only for inequality constraints (equalities handled separately), logarithm undefined at boundary.
Projected gradient descent: Iterates always feasible, simple updates, no parameter tuning (ρ or t). Cons: requires efficient projection onto C, only practical when P_C has a closed form or is cheap to compute, slower convergence.
Rule of thumb: Use projected GD when C has an easy projection (box, ball, simplex). Use barrier method for general smooth inequality constraints with moderate m. Use penalty method as a quick prototyping tool or when the other methods are impractical.

Mixed Challenging Exercises

Exercise 6.1 [Hard]
Consider training a linear classifier with a margin constraint. Given data points (a_i, y_i) with $a_i \in \mathbb{R}^n$ and $y_i \in \{-1, +1\}$ for $i = 1, \ldots, m$, we solve:

$$\min_{w \in \mathbb{R}^n} \frac{1}{2}\|w\|_2^2 \quad \text{s.t.} \quad y_i(a_i^T w) \geq 1, \; i = 1, \ldots, m.$$

(a) This is a quadratic program with linear inequality constraints. Write the Lagrangian and derive the KKT conditions.

(b) Formulate the barrier subproblem for parameter $t > 0$. Write the gradient of the barrier objective.

(c) Alternatively, one could solve this via projected gradient descent. The constraint set is $C = \{w : y_i a_i^T w \geq 1, \forall i\}$. Is C convex? Is the projection onto C easy to compute in general?

(d) Based on your answers, argue which method (barrier vs. projected GD vs. penalty) is most appropriate for this problem.

Solution

(a) Rewrite constraints as $g_i(w) = 1 - y_i a_i^T w \leq 0$.
Lagrangian: $\mathcal{L}(w, \alpha) = \frac{1}{2}\|w\|^2 + \sum_{i=1}^m \alpha_i(1 - y_i a_i^T w)$.
KKT conditions:

1. Stationarity: $w - \sum_i \alpha_i y_i a_i = 0 \implies w = \sum_i \alpha_i y_i a_i$.
2. Primal feasibility: $y_i a_i^T w \geq 1$ for all i.
3. Dual feasibility: $\alpha_i \geq 0$ for all i.
4. Complementary slackness: $\alpha_i(1 - y_i a_i^T w) = 0$ for all i.

From stationarity, w is a linear combination of the data points (the support vectors are those with $\alpha_i > 0$).
(b) Barrier subproblem:

$$\min_w \frac{t}{2}\|w\|^2 - \sum_{i=1}^m \log(y_i a_i^T w - 1).$$

Gradient:

$$\nabla_w B_t(w) = tw + \sum_{i=1}^m \frac{-y_i a_i}{y_i a_i^T w - 1} = tw - \sum_{i=1}^m \frac{y_i a_i}{y_i a_i^T w - 1}.$$

(c) C is convex: it is the intersection of half-spaces $\{w : y_i a_i^T w \geq 1\}$, and the intersection of convex sets is convex.
However, the projection onto C (intersection of m half-spaces, i.e. a polyhedron) does not have a simple closed form in general. Computing $P_C(z)$ itself requires solving a QP: $\min_{w \in C} \|w - z\|^2$. This makes projected gradient descent impractical — each step requires solving a problem of comparable difficulty.
(d) *Barrier method* is most appropriate:

- The constraints are smooth linear inequalities, so the barrier function is well-defined.
- Each centering step is an unconstrained smooth problem solvable by Newton's method.
- The duality gap m/t gives a clear stopping criterion.

Projected GD is impractical due to the expensive projection. The penalty method would work but offers weaker convergence guarantees and produces infeasible iterates.

Exercise 6.2 [Hard]
Consider the entropy-regularized optimal transport problem in 1D. Given two discrete distributions $p, q \in \mathbb{R}^n$ on the probability simplex, and a cost matrix $C \in \mathbb{R}^{n \times n}$, we solve:

$$\min_{X \in \mathbb{R}^{n \times n}} \langle C, X \rangle + \frac{1}{t} \sum_{i,j} X_{ij} \log X_{ij}$$

$$\text{s.t.} \quad X\mathbf{1} = p, \quad X^T\mathbf{1} = q, \quad X_{ij} \geq 0.$$

(a) Show that the objective is strictly convex in X for $t > 0$ and $X_{ij} > 0$.

(b) Write the Lagrangian (ignoring the non-negativity constraints, as the entropy term ensures $X_{ij} > 0$ at the optimum). From the KKT stationarity condition, show that the optimal X^* has the form $X^*_{ij} = u_i K_{ij} v_j$ for some vectors $u, v > 0$ and matrix $K_{ij} = e^{-tC_{ij}}$.

(c) The problem in (b) can be seen as the equality-constrained optimization on the positive orthant. Explain why the barrier method is unnecessary here — the entropy regularization already acts as a barrier.

(d) If instead we dropped the entropy term and solved the pure LP with the simplex constraints, explain how the interior point method would approach this problem.

Solution

(a) The Hessian of $f(X) = \sum_{ij} X_{ij} \log X_{ij}$ with respect to X (viewed as an n^2-vector) is diagonal with entries $1/X_{ij} > 0$ for $X_{ij} > 0$. Thus f is strictly convex. The linear term $\langle C, X\rangle$ is convex (affine). The sum $\langle C, X\rangle + \frac{1}{t}\sum X_{ij} \log X_{ij}$ is strictly convex for $t > 0$.
(b) Lagrangian (with multipliers $\alpha \in \mathbb{R}^n$ for row sums, $\beta \in \mathbb{R}^n$ for column sums):

$$\mathcal{L} = \sum_{ij} C_{ij} X_{ij} + \frac{1}{t}\sum_{ij} X_{ij} \log X_{ij} + \alpha^T(X\mathbf{1} - p) + \beta^T(X^T\mathbf{1} - q).$$

Stationarity w.r.t. X_{ij}:

$$C_{ij} + \frac{1}{t}(\log X_{ij} + 1) + \alpha_i + \beta_j = 0.$$

$$\log X_{ij} = -tC_{ij} - t\alpha_i - t\beta_j - 1.$$

$$X_{ij} = e^{-t\alpha_i - 1/2} \cdot e^{-tC_{ij}} \cdot e^{-t\beta_j - 1/2} = u_i K_{ij} v_j,$$

where $u_i = e^{-t\alpha_i - 1/2}$, $v_j = e^{-t\beta_j - 1/2}$, $K_{ij} = e^{-tC_{ij}}$. Since exponentials are positive, $u, v > 0$.
(c) The entropy term $\frac{1}{t}\sum X_{ij} \log X_{ij}$ diverges to $+\infty$ as any $X_{ij} \to 0^+$ (since $x \log x \to 0$ but the gradient $\log x + 1 \to -\infty$, keeping the optimum in the interior). This plays the same role as a barrier: it automatically prevents the solution from hitting the boundary $X_{ij} = 0$. Thus, the non-negativity constraints are automatically satisfied at the optimum and we only need to handle the equality constraints.
(d) Without entropy regularization, the problem is a linear program: $\min\langle C, X\rangle$ s.t. $X\mathbf{1} = p$, $X^T\mathbf{1} = q$, $X \geq 0$.
The interior point method would:

1. Add the logarithmic barrier: $-\sum_{ij} \log(X_{ij})$ for the non-negativity constraints.
2. Solve the sequence: $\min_X t\langle C, X\rangle - \sum_{ij} \log(X_{ij})$ s.t. $X\mathbf{1} = p$, $X^T\mathbf{1} = q$.
3. Increase t by factor γ and re-solve, using the previous solution as warm start.
4. Stop when $m/t = n^2/t < \varepsilon$.

Note the structural similarity: the barrier subproblem at step 2 is exactly the entropy-regularized problem (with the identification $1/t_{\text{barrier}} \leftrightarrow 1/t_{\text{entropy}}$). Thus, entropy-regularized OT can be viewed as solving one centering step of the barrier method.

Exercise 6.3 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be μ-strongly convex with L-Lipschitz continuous gradient, and let C be a closed convex set.

(a) Show that the constrained problem $\min_{x \in C} f(x)$ has a unique minimizer x^*.

(b) For projected gradient descent with step size $t = 1/L$, prove the descent lemma:

$$\|x^{(k+1)} - x^*\|^2 \leq \left(1 - \frac{\mu}{L}\right) \|x^{(k)} - x^*\|^2.$$

(Hint: Use strong convexity, the Lipschitz gradient condition, and the non-expansiveness of P_C.)

(c) How many iterations does projected gradient descent need to achieve $\|x^{(k)} - x^*\|^2 \leq \varepsilon$ as a function of $\kappa = L/\mu$ (the condition number)? Compare with the accelerated rate.

(d) Relate the condition number κ to a concrete ML setting: for Ridge regression $\min_{\|\theta\|_1 \leq \tau} \frac{1}{2}\|X\theta - y\|^2 + \frac{\alpha}{2}\|\theta\|^2$, what is κ?

Solution

(a) By strong convexity, for any $x, y \in C$ and $\lambda \in (0,1)$:

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y) - \frac{\mu}{2}\lambda(1-\lambda)\|x-y\|^2.$$

Strict inequality when $x \neq y$, so f is strictly convex on C. Since f is continuous, coercive (grows as $\|x\| \to \infty$), and C is closed, a minimizer exists. Strict convexity guarantees uniqueness.

(b) The gradient step gives $z^{(k)} = x^{(k)} - \frac{1}{L}\nabla f(x^{(k)})$, and $x^{(k+1)} = P_C(z^{(k)})$.
Since $x^* = P_C(x^* - \frac{1}{L}\nabla f(x^*))$ (optimality condition), by non-expansiveness:

$$\|x^{(k+1)} - x^*\|^2 \leq \|z^{(k)} - (x^* - \frac{1}{L}\nabla f(x^*))\|^2.$$

$$= \|(x^{(k)} - x^*) - \frac{1}{L}(\nabla f(x^{(k)}) - \nabla f(x^*))\|^2.$$

Let $\delta = x^{(k)} - x^*$ and $g = \nabla f(x^{(k)}) - \nabla f(x^*)$. Then:

$$\|\delta - g/L\|^2 = \|\delta\|^2 - \frac{2}{L}\langle \delta, g\rangle + \frac{1}{L^2}\|g\|^2.$$

Using co-coercivity (from μ-strong convexity and L-smoothness): $\langle \delta, g\rangle \geq \frac{\mu L}{\mu+L}\|\delta\|^2 + \frac{1}{\mu+L}\|g\|^2$.
And $\|g\|^2 \leq L^2\|\delta\|^2$ from Lipschitz. Using the simpler bound $\langle \delta, g\rangle \geq \mu\|\delta\|^2$ (from strong convexity):

$$\|x^{(k+1)} - x^*\|^2 \leq \|\delta\|^2 - \frac{2\mu}{L}\|\delta\|^2 + \frac{L^2\|\delta\|^2}{L^2} = (1 - \frac{2\mu}{L} + 1)\|\delta\|^2.$$

This is too loose. Using the tighter co-coercivity bound or the standard result from L-smoothness that $\frac{1}{L}\|g\|^2 \leq \langle \delta, g\rangle$:

$$\|\delta - g/L\|^2 = \|\delta\|^2 - \frac{2}{L}\langle \delta, g\rangle + \frac{\|g\|^2}{L^2} \leq \|\delta\|^2 - \frac{1}{L}\langle \delta, g\rangle.$$

Now using $\langle \delta, g\rangle \geq \mu\|\delta\|^2$: $\leq (1 - \mu/L)\|\delta\|^2$.

(c) From $\|x^{(k)} - x^*\|^2 \leq (1 - 1/\kappa)^k\|x^{(0)} - x^*\|^2$, to achieve ε:

$$(1 - 1/\kappa)^k \leq \varepsilon/\|x^{(0)} - x^*\|^2 \implies k \geq \kappa \log(\|x^{(0)} - x^*\|^2/\varepsilon).$$

So $O(\kappa \log(1/\varepsilon))$ iterations.
Accelerated PGD: $O(\sqrt{\kappa}\log(1/\varepsilon))$ iterations, a square-root improvement.

(d) The objective is $f(\theta) = \frac{1}{2}\|X\theta - y\|^2 + \frac{\alpha}{2}\|\theta\|^2$, so $\nabla^2 f = X^T X + \alpha I$.
$L = \lambda_{\max}(X^T X) + \alpha = \sigma_{\max}(X)^2 + \alpha$. $\mu = \lambda_{\min}(X^T X) + \alpha = \sigma_{\min}(X)^2 + \alpha$.

$$\kappa = \frac{\sigma_{\max}(X)^2 + \alpha}{\sigma_{\min}(X)^2 + \alpha}.$$

The regularization α improves conditioning: without it ($\alpha = 0$), κ could be infinite if X is rank-deficient. With $\alpha > 0$, κ is always finite.